

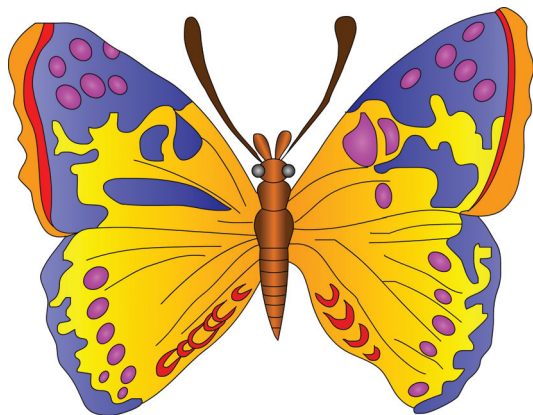


Fig. 3.25 Butterfly

Class-iv: Insecta

Insecta includes the most successful and the largest group of animals on the Earth. They are cosmopolitan, but absent from the marine habitat. Body consists of head, thorax and abdomen. Head bears paired compound eyes, antennae, mandibles, maxillae and an unpaired labium. Thorax consists of three segments and each segment bears a pair of *jointed walking legs*, hence this class is also referred to as **hexapoda**; thorax also bears one to two pairs of wings or wings may be absent (*Lepisma*). Respiratory organs are tracheae. Excretory organs are **Malpighian tubules**; they are **uricotelic** (excrete uric acid crystals as a **water conservation adaptation**). Development is mostly **indirect** with larval stages and **metamorphosis**.

Examples: *Periplaneta* (cockroach), *Musca* (house fly), *Lepisma* (silver fish).

3.8 PHYLUM-MOLLUSCA

- Unsegmented schizocoelomates

3.8.1 General Characters

Mollusca is the **second largest** phylum (Figures 3.26-3.28) in the Kingdom Animalia. Molluscs are terrestrial or aquatic (marine or fresh water). They are **bilaterally symmetrical** (some are asymmetrical, e.g. snail) and **coelomate** animals. Body is covered by a **calcareous shell** (mostly external, in some internal or reduced or absent) and is **un-segmented** with a distinct **head**, **muscular foot** and **visceral mass / visceral hump**. A soft and spongy layer of skin forms the **mantle** (*pallium*) over the visceral mass and secretes the shell. The space between the visceral mass and the mantle is called the '*mantle cavity*' in which comb like gills (*ctenidia*) are present. They have

Fig. 3.26 *Pila globosa*

respiratory and excretory functions (Ref: NCERT text book). The anterior head region has sensory tentacles. The body cavity is a **haemocoel**, and the true coelom is confined to the kidneys, gonads and pericardial space. The buccal cavity contains a file-like rasping organ called **radula** for feeding except in the bivalves. Except for cephalopods circulatory system is of *open type*; haemolymph contains a copper containing respiratory pigment, the **haemocyanin**. Nervous system consists of ganglia, commissures and connectives; sense organs are tentacles, eyes and **osphradium** (tests the purity of water – present in the bivalves and gastropods). Excretory organs are

metanephridia (commonly called *kidneys* or renal organs). They are usually dioecious (unisexual /gonochoric) and oviparous with indirect development which includes a characteristic ciliated **veliger larva** (a modified trochophore).

Examples: *Pila*, *Pinctada*, *Sepia*, *Loligo*, *Octopus*, *Aplysia*, *Dentalium*.

3.8.2 Classification of the phylum Mollusca

Phylum Mollusca is classified into seven classes, namely **1. Aplacophora** **2. Polyplacophora** **3. Monoplacophora** **4. Gastropoda** **5. Scaphopoda** **6. Pelecypoda** and **7. Cephalopoda**.

Class-1: Aplacophora (Gr. *a* : without; *placos* : plate; *pherein* : bearing)

They are primitive 'worm like' marine molluscs without mantle, shell, foot and nephridia. Head is poorly developed; a **radula** is present. Cuticle contains **calcareous spicules**. In some there is a *mid-ventral groove* which is homologous to the foot of the other molluscs.

Examples: *Neomenia*, *Chaetoderma*.

Class-2: Polyplacophora (Gr. *poly* : many; *placos* : plate; *pherein* : bearing)

This class includes **chitons**. They are bilaterally symmetrical and dorsoventrally flattened. Shell is dorsal and consists of **eight** transverse plates (*valves*). Foot is ventral, elongated and flat. Gills are 6 to 88 pairs. Development includes a trochophore larva.

Examples: *Chiton*, *Lepidopleurus*.

Class-3: Monoplacophora (Gr. *Mono* : single; *placos* : plate; *pherein* : bearing)

The members of this class were believed to have become extinct millions of years ago; *Neopilina* was collected from the deep sea off the Pacific coast of Costa Rica in 1952 by the oceanographic research vessel called **Galathea**. They are bilaterally symmetrical. Nephridia and gills show '**serial repetition**' (some call it **internal segmentation**). Shell is single and plate like. Heart is unique with two pairs of atria opening into two ventricles.

Example: *Neopilina galathea*.

Class-4: Gastropoda (Gr. *Gaster* : belly; *podos* : foot)

Gastropoda is the largest and most diverse class of the phylum Mollusca; includes snails, slugs, limpets, etc.; mostly marine; some are freshwater or terrestrial. Shell is external, univalve and spirally coiled or internal (*Aplysia*) or absent (*Doris*). Head is distinct with eyes and tentacles. **Radula** occurs in the buccal cavity. Foot is ventral, flattened, muscular and helps in creeping. Respiratory organs are gills or lungs (pulmonary sacs) or both. During embryonic development, one side of the visceral mass grows faster than the other side; this uneven growth rotates the visceral mass up to 180°. This process is called '**torsion**'. They are mostly dioecious; development is indirect and includes **veliger larva**.

Examples: *Pila* (apple snail), *Aplysia* (sea hare), *Doris* (sea lemon).



Fig. 3.27 *Pila globosa* (shell)

Class-5: Scaphopoda (Gr. *Scapho* : boat, hollow shell; *podos* : foot)

They are marine and are commonly called '**tusk shells**' or '**tooth shells**' as the shell is tubular and '*tusk*' like, opening at both the ends. Foot is *cone* like and useful for digging. Eyes, tentacles, *ctenidia* (gills) and atria are absent. A large number of thread like structures called **captacula** help in the capture of food.

Examples: *Dentalium*, *Pulsellum*.

Class-6: Pelecypoda or Bivalvia or Lamellibranchiata (Gr. *pelekys* : axe, hatchet; *podos* : foot)

This class includes *mussels*, *oysters*, *clams*, etc. Body is laterally compressed and covered by a shell made up of two valves (Bivalvia). Head and sensory appendages are greatly reduced. Foot is '*wedge shaped*' (hence the name pelecypoda) and adapted for digging. Byssus threads occur in some (e.g. *Mytilus*) and help in attaching to the substratum. Radula is absent. They are '*suspension feeders*' or '*filter feeders*'. In some a '**crystalline style**' occurs in the stomach and helps in the digestion of starches. Respiratory organs are '*plate like*' gills (hence the name lamellibranchiata). They are dioecious (sexes are separate); development is indirect including larval forms such as **trochophore** and **veliger**; a **glochidium larva** occurs in freshwater mussels such as *Unio* (adapted for parasitic life on the gills of fish).

Examples: *Unio* (freshwater mussel), *Mytilus* (marine mussel), *Pinctada* (pearl oyster).

Class-7: Cephalopoda or Siphonopoda (Gr. *cephalo* : head; *podos* : foot)

This class includes *cuttle fishes*, *squids*, *octopuses*, *nautili* etc. Head is distinct with conspicuous eyes similar to those of vertebrates, a pair of horny beak like jaws and a radula occur in the buccal cavity. Shell may be external and multi-chambered (*Nautilus*) or internal (*Sepia*, *Loligo*) or absent (*Octopus*). The shell of *Sepia* is commonly called '**cuttle bone**' and that of *Loligo* is commonly called '**pen**'. Foot is modified into eight (*octopus*) or ten arms (*Sepia*, *Loligo*) provided with suckers present around the mouth and a part of the foot is modified into a '**siphon**' (useful in swift darting movements). Some possess **ink gland** and eject a cloud of ink to escape from the



Fig. 3.28 Octopus

predators (*defensive adaptation*). Ctenidia, atria and nephridia are *two* in **dibranchiates** (*Sepia*) and *four* in **tetrabranchiates** (*Nautilus*). Circulatory system is a **closed type** (*unique feature of cephalopoda*); heart has two to four atria and a ventricle. Nervous system is well developed with a well developed brain enclosed in a **cartilaginous cranium** (*brain case*). They are dioecious (sexes are separate); development is direct.

Examples: *Sepia* (cuttle fish), *Architeuthis* (giant squid - the largest living invertebrate), *Nautilus*, *Octopus* (devil fish).